
Modbus Interface Specifications for Exicom Hybrid Power Plants Using Controller M2000

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1 Introduction

This document explains the implementation of the Modbus communications protocol specific to Exicom Hybrid Power Plants. The Modbus protocol is an industry standard that allows a master station to interrogate devices for its publically available data and set supported control and configuration parameters.

This document assumes the reader is already familiar with the Modbus protocol and serial communication. Kindly refer to Modbus IDA document “Modbus_Application_Protocol_V1_1b” at <http://www.Modbus-IDA.org> for full details of the Modbus specifications.

2 Modbus Configuration

2.1 Supported Modes of Transmission

The Modbus protocol supports two serial transmission modes, ASCII and RTU (Remote Terminal Unit). Exicom products support the RTU mode of operation, which is more prevalent.

2.2 Communications Interface

Exicom Hybrid Power Products support RS 485, 2Wire interface for Modbus communications.

2.3 Communications Parameters

Serial communications parameters are set to 19200B, 8 data bits, No parity and 1 Stop bit.

3 Supported Modbus Functions

Exicom Hybrid Power Plants support following Function codes over Modbus I/f. All the communications uses Modbus CRC as specified in the recommendations.

3.1 Function 4 (04h): Read Input Data Registers

This function code is used to read the contents of one or more Input data registers on the selected slave.

3.1.1 Query

The query message specifies the starting register and the number of registers to read.

Table 1 - Function 04 Query Message Format

Slave Addr	Function Code 04	Starting Register High	Starting Register Low	# of Register High	# of Register Low	CRC Low	CRC High
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3.1.2 Response

The response message contains the data read from registers on the slave. The registers occur in order from the first register requested through each sequential register that follows for the number of registers requested.

Table 2- Function 04 Response Message Format

Slave Addr	Function Code 04	Byte Count	Register Data Val High	Register Data Val Low	...	CRC Low	CRC High
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4 Slave Address Assignment

Before starting any communications with Exicom device over Modbus, Slave ID for device should be set from front panel. This can be set at the time of installation or commissioning. Allowable range for Slave ID is from 1-247.

Default Slave ID is “1”.

5 Modbus Registers Data Types

5.1 Register Format

Modbus Registers are always 16 bit wide. Different data types are mapped to Modbus Register as detailed below.

5.1.1 Packed Boolean Format - Bool

Boolean, or two state values, are packed 16 per register. Individual bits are referenced in the address map according to their bit number within the register, B0 through B15.

Table 3 - Packed Boolean Format

Register N
B15 B14 B13 B12 B11 B10 B9 B8 B7 B6 B5 B4 B3 B2 B1 B0

5.1.2 8-bit Integer Format – Uint8 / Sint8

8-bit values are packed to a register, with high byte set as 00 and lower byte field holds the value. For such variable High byte is always 0.

Table 4 - Packed 8-bit Integer Format

<i>Register N</i>	
<i>High Byte</i>	<i>Low Byte</i>
<i>B7 B0</i>	<i>B7 B0</i>

5.1.3 Strings

If the format is used for string data, then a fixed range of contiguous registers are specified along with an indication of how many bytes are contained in the string. Data is arranged such that the start of the string begins in the high byte of register N and the last byte of the string occupies the low byte of register $N + [(x - 1) / 2]$, where x is the number of bytes in the string. Strings are “nul” terminated, and are “nul” padded to fill the remaining byte(s) in the register(s). String sizes exceeding the total payload length of a Modbus PDU are not permitted.

Table 5 - String Format

Register N		Register N+1		Register ...		Register N+[(x-1)/2]	
Char	Char	Char	Char	Char	Char	Char	Char
C1	C2	C3	C4	C...	C...	Cx-1	Cx

5.1.4 16-bit Integer Format – Uint16 and Sint16

16-bit integer values are contained in a single register. If the value is unsigned then the range of valid values is 0 to 65535. If the value is signed then the range is -32768 to +32767 in two's complement. Note that in each case the maximum value is limited by required reserved values.

Table 6 - 16-bit Integer Format

Register N
B15 B8 B7 B0

The signed and unsigned versions of this format are referred to in the register map as Sint16 and Uint16, respectively. Byte order in the register is Big Endian i.e. High order byte maps to B15-B8 and low order byte maps to B7-B0.

5.1.5 32-bit Integer Format – Uint32 and Sint32

To accommodate values larger than that reached with a 16-bit number, a 32-bit format is provided that spans two registers. In signed and unsigned 32 bit integer formats the 32-bit value is split between two consecutive 16-bit registers. The first register (at address N) is the high-order word, and the second register (at address N + 1) is the low-order word:

$$\text{Value} = \text{Value}(\text{registerN}) * 65535 + \text{Value}(\text{registerN}+1)$$

For all read and write operations on 32-bit integers, both the registers should be read / written.

Table 7 - 32-bit Integer Format

Register N	Register N+1
B31 B23 B22 B16	B15 B8 B7 B0

If the value is unsigned then the range of valid values is 0 to 4,294,967,295. If the value is signed then the range is -2,147,483,648 to +2,147,483,647 in two's complement. The signed and unsigned versions of this format are referred to in the register map as Sint32 and Uint32, respectively. The overall byte order for both registers is Big Endian.

5.2 Numeric Units and Scale

5.2.1 Units

Scalar values are generally useless without some indication of the units of measure. The units for each register are noted in the register map. Where the units are left blank, the associated value is unit-less, such as a count or enumeration.

5.2.2 Scale

Numeric values are scaled to represent real numbers in an integer format. This fixed-point representation is fixed for each register. For example, a battery voltage of 48.4 VDC would be represented with an integer value of 0484 using X10 scaling. The scaling of each register is noted in the register map. Where the scale is left blank, X1 scaling is assumed or the value is scale-less, such as a count or enumeration.

6 Register Access & Exceptions

Exicom only employs the Modbus Input Registers & Holding Registers. By definition all Input Registers are read only and Holding Registers are read-write. Depending on the operation, the device may restrict access to certain registers to read-only. Table 10 lists Register details for Input & Holding Registers. Note that in these tables Modbus Register Numbers and Modbus Register Address both are mentioned. Modbus Registers Numbers are generally appended with a prefix indicating type of register. Following conventions are used

Input Registers Numbers : 4xxxx

Holding Register Numbers : 3xxxx

However note that in the message format only Register Address is sent, which starts from 0000 for both Input Registers & Holding Registers. The device differentiates the table accessed from Function Code.

For both these tables' data can be read in a single message only up to maximum of limits defined in Modbus Protocol Ver1.1b or table length as specified in Table 10.

Any access of parameters not in line with above restrictions will return an exception code and no action shall be taken. All the Exception codes returned by Exicom devices are mentioned in Table 9.

In case an invalid data (out of limit) is written to a Holding register, an exception will be returned. However for multiple register write operation, all registers prior to the register having invalid data value will be written. All registers in write command after and including the register for which invalid value is specified will be left untouched.

Table 8 below specifies the register access limits:

Table 8- Register Access Limits

Function Code	Maximum Number of Registers accessed in single command as per Modbus	Maximum Number of Registers accessed in single command as per Device	Limit
04, Read Input Registers	125	Last accessed register should be available in Table 10.	Minimum of the two
03, Read Holding Register	125		
06, Write Single Holding Register	1		1
01, Read Coil			

6.1 Exception Responses

Exceptions may be generated in response to commands from the Modbus master to signify reasons why a request packet cannot be honored. The table below describes the exception codes supported by Exicom devices along with their possible causes.

Table 9 – Exception Codes

Code	Name	Meaning
01	Illegal Function	The function code received in the query is not an allowable action for the device. Exicom devices only support functions 3, 4, 6 and 16.
02	Illegal Address	The address referenced in the data field of the request packet is invalid for the specified function.
03	Illegal Value	This indicates a fault in the structure of request, such as that the implied length is incorrect. It specifically does NOT mean that a data item submitted for storage in a register has a value outside the expectation of the device.
04	SLAVE DEVICE FAILURE	An unrecoverable error occurred while the device was attempting to perform the requested action. OR The value referenced in the data field of the request packet is not allowed for the referenced register.

6.2 Read-Only Input Registers

All alarm status & performance parameters are available as Input Registers. The Data Type, scaling factor etc. are also specified Input Register details. For reading Input Registers, Function Code 04 shall be used. Input Registers details are given in Table 10.

6.3 Invalid Registers

In the register map, there are no gaps between registers. Access to invalid registers will return appropriate exception code as per Table 9.

7 Modbus Register Maps

7.1 Input Registers

Modbus address for read input register (read only).

Table 10 - Input Register Map

Reg. Address	Parameter	Data Type	Units	Remarks
0	Alarm Status Reg. 1	BOOL		See Table 11 for details
1	Alarm Status Reg. 2	BOOL		See Table 11 for details
2	Alarm Status Reg. 3	BOOL		See Table 11 for details
3	Alarm Status Reg. 4	BOOL		See Table 11 for details
4	Rectifier Alarm Status	BOOL		See Table 11 for details
5	Rectifier Communication Status	BOOL		See Table 11 for details
6	Solar Alarm Status	BOOL		See Table 11 for details
7	Solar Communication Status	BOOL		See Table 11 for details
8	Rectifier Bus Voltage	UINT16	Volt	Divide by 10 to get value
9	System Current	SINT16	Amp	
10	Load Voltage	UINT16	Volt	Divide by 10 to get value
11	Load Current	SINT16	Amp	Divide by 10 to get value
12	Battery 1 Voltage	UINT16	Volt	Divide by 10 to get value
13	Battery 1 Current	SINT16	Amp	Divide by 10 to get value
14	Battery 1 Remaining AH %age	UINT16	%	Divide by 10 to get value
15	Battery 2 Voltage	UINT16	Volt	Divide by 10 to get value
16	Battery 2 Current	SINT16	Amp	Divide by 10 to get value
17	Battery 2 Remaining AH %age	UINT16	%	Divide by 10 to get value
18	Line 1 Voltage	UINT16	Volt	
19	Line 2 Voltage	UINT16	Volt	
20	Line 3 Voltage	UINT16	Volt	
21	Battery Temperature	UINT16	'C	Divide by 10 to get value
22	Reserved	UINT16		
23	Reserved	UINT16		
24	Output Voltage Rectifier [1]	UINT16	Volt	Divide by 10 to get value
25	Output Voltage Rectifier [2]	UINT16	Volt	Divide by 10 to get value
26	Output Voltage Rectifier [3]	UINT16	Volt	Divide by 10 to get value
27	Output Voltage Rectifier [4]	UINT16	Volt	Divide by 10 to get value
28	Output Voltage Rectifier [5]	UINT16	Volt	Divide by 10 to get value
29	Output Voltage Rectifier [6]	UINT16	Volt	Divide by 10 to get value
30	Output Voltage Rectifier [7]	UINT16	Volt	Divide by 10 to get value
31	Output Voltage Rectifier [8]	UINT16	Volt	Divide by 10 to get value
32	Output Voltage Rectifier [9]	UINT16	Volt	Divide by 10 to get value
33	Output Voltage Rectifier [10]	UINT16	Volt	Divide by 10 to get value
34	Output Voltage Rectifier [11]	UINT16	Volt	Divide by 10 to get value
35	Output Voltage Rectifier [12]	UINT16	Volt	Divide by 10 to get value
36	Output Voltage Rectifier [13]	UINT16	Volt	Divide by 10 to get value

Reg. Address	Parameter	Data Type	Units	Remarks
37	Output Voltage Rectifier [14]	UINT16	Volt	Divide by 10 to get value
38	Output Voltage Rectifier [15]	UINT16	Volt	Divide by 10 to get value
39	Output Voltage Rectifier [16]	UINT16	Volt	Divide by 10 to get value
40	Output Current Rectifier [1]	UINT16	Amp	Divide by 10 to get value
41	Output Current Rectifier [2]	UINT16	Amp	Divide by 10 to get value
42	Output Current Rectifier [3]	UINT16	Amp	Divide by 10 to get value
43	Output Current Rectifier [4]	UINT16	Amp	Divide by 10 to get value
44	Output Current Rectifier [5]	UINT16	Amp	Divide by 10 to get value
45	Output Current Rectifier [6]	UINT16	Amp	Divide by 10 to get value
46	Output Current Rectifier [7]	UINT16	Amp	Divide by 10 to get value
47	Output Current Rectifier [8]	UINT16	Amp	Divide by 10 to get value
48	Output Current Rectifier [9]	UINT16	Amp	Divide by 10 to get value
49	Output Current Rectifier [10]	UINT16	Amp	Divide by 10 to get value
50	Output Current Rectifier [11]	UINT16	Amp	Divide by 10 to get value
51	Output Current Rectifier [12]	UINT16	Amp	Divide by 10 to get value
52	Output Current Rectifier [13]	UINT16	Amp	Divide by 10 to get value
53	Output Current Rectifier [14]	UINT16	Amp	Divide by 10 to get value
54	Output Current Rectifier [15]	UINT16	Amp	Divide by 10 to get value
55	Output Current Rectifier [16]	UINT16	Amp	Divide by 10 to get value
56	Rectifier Load Share Deviation %age	UINT16	%	
57	Rectifier Current Sum	UINT32	Amp	Divide by 10 to get value
58	Rectifier Current Sum	UINT32	Amp	Divide by 10 to get value
59	Active Rectifier Count	UINT8		
60	Reserved	UINT16		
61	Reserved	UINT16		
62	Reserved	UINT16		
63	Reserved	UINT16		
64	Reserved	UINT16		
65	Reserved	UINT16		
65	Reserved	UINT16		
66	Reserved	UINT16		
67	Reserved	UINT16		
68	Reserved	UINT16		
69	Reserved	UINT16		
70	Reserved	UINT16		
71	Reserved	UINT16		
72	Reserved	UINT16		
73	Reserved	UINT16		
74	Reserved	UINT16		
75	Reserved	UINT16		
76	Reserved	UINT16		
77	Reserved	UINT16		
78	Reserved	UINT16		

Reg. Address	Parameter	Data Type	Units	Remarks
79	Reserved	UINT16		
80	Reserved	UINT16		
81	Operator Power[1]	UINT32	Watt	
82	Operator Power[1]	UINT32	Watt	
83	Operator Cumulative Power[1]	UINT32	Watt	
84	Operator Cumulative Power[1]	UINT32	Watt	
85	Battery Charge Counter	UINT32		
86	Battery Charge Counter	UINT32		
87	Battery Discharge Counter	UINT32		
88	Battery Discharge Counter	UINT32		
89	Battery Cumulative Charge Time	UINT32	Hour	Divide by 10 to get value
90	Battery Cumulative Charge Time	UINT32	Hour	Divide by 10 to get value
91	Battery Cumulative Discharge Time	UINT32	Hour	Divide by 10 to get value
92	Battery Cumulative Discharge Time	UINT32	Hour	Divide by 10 to get value
93	Battery Cumulative Charge Power	UINT32	KWH	Divide by 10 to get value
94	Battery Cumulative Charge Power	UINT32	KWH	Divide by 10 to get value
95	Battery Cumulative Discharge Power	UINT32	KWH	Divide by 10 to get value
96	Battery Cumulative Discharge Power	UINT32	KWH	Divide by 10 to get value
97	Battery Cumulative Discharge AH	UINT32	KAH	Divide by 10 to get value
98	Battery Cumulative Discharge AH	UINT32	KAH	Divide by 10 to get value
99	Total Solar Cumulative Time	UINT32	Hour	Divide by 10 to get value
100	Total Solar Cumulative Time	UINT32	Hour	Divide by 10 to get value
101	Total Solar Cumulative Power	UINT32	KWH	Divide by 10 to get value
102	Total Solar Cumulative Power	UINT32	KWH	Divide by 10 to get value
103	Solar Rectifier Input Voltage [1]	UINT16	Volt	Divide by 10 to get value
104	Solar Rectifier Input Voltage [2]	UINT16	Volt	Divide by 10 to get value
105	Solar Rectifier Input Voltage [3]	UINT16	Volt	Divide by 10 to get value
106	Solar Rectifier Input Voltage [4]	Unit16	Volt	Divide by 10 to get value
107	Solar Rectifier Input Voltage [5]	Unit16	Volt	Divide by 10 to get value
108	Solar Rectifier Input Voltage [6]	Unit16	Volt	Divide by 10 to get value
109	Solar Rectifier Input Voltage [7]	Unit16	Volt	Divide by 10 to get value
110	Solar Rectifier Input Voltage [8]	Unit16	Volt	Divide by 10 to get value
111	Solar Rectifier Input Voltage [9]	Unit16	Volt	Divide by 10 to get value
112	Solar Rectifier Input Voltage [10]	Unit16	Volt	Divide by 10 to get value
113	Solar Rectifier Input Current [1]	Unit16	Amp	Divide by 10 to get value
114	Solar Rectifier Input Current [2]	Unit16	Amp	Divide by 10 to get value
115	Solar Rectifier Input Current [3]	Unit16	Amp	Divide by 10 to get value
116	Solar Rectifier Input Current [4]	Unit16	Amp	Divide by 10 to get value
117	Solar Rectifier Input Current [5]	Unit16	Amp	Divide by 10 to get value
118	Solar Rectifier Input Current [6]	Unit16	Amp	Divide by 10 to get value
119	Solar Rectifier Input Current [7]	Unit16	Amp	Divide by 10 to get value

Reg. Address	Parameter	Data Type	Units	Remarks
120	Solar Rectifier Input Current [8]	Unit16	Amp	Divide by 10 to get value
121	Solar Rectifier Input Current [9]	Unit16	Amp	Divide by 10 to get value
122	Solar Rectifier Input Current [10]	Unit16	Amp	Divide by 10 to get value
123	Solar Rectifier Input Power[1]	Unit32	Watt	
124	Solar Rectifier Input Power[1]	Unit32	Watt	
125	Solar Rectifier Input Power[2]	Unit32	Watt	
126	Solar Rectifier Input Power[2]	Unit32	Watt	
127	Solar Rectifier Input Power[3]	Unit32	Watt	
128	Solar Rectifier Input Power[3]	Unit32	Watt	
129	Solar Rectifier Input Power[4]	Unit32	Watt	
130	Solar Rectifier Input Power[4]	Unit32	Watt	
131	Solar Rectifier Input Power[5]	Unit32	Watt	
132	Solar Rectifier Input Power[5]	Unit32	Watt	
133	Solar Rectifier Input Power[6]	Unit32	Watt	
134	Solar Rectifier Input Power[6]	Unit32	Watt	
135	Solar Rectifier Input Power[7]	Unit32	Watt	
136	Solar Rectifier Input Power[7]	Unit32	Watt	
137	Solar Rectifier Input Power[8]	Unit32	Watt	
138	Solar Rectifier Input Power[8]	Unit32	Watt	
139	Solar Rectifier Input Power[9]	Unit32	Watt	
140	Solar Rectifier Input Power[9]	Unit32	Watt	
141	Solar Rectifier Input Power[10]	Unit32	Watt	
142	Solar Rectifier Input Power[10]	Unit32	Watt	
143	Solar Rectifier Total Power[1]	Unit32	KWH	Divide by 10 to get value
144	Solar Rectifier Total Power[1]	Unit32	KWH	Divide by 10 to get value
145	Solar Rectifier Total Power[2]	Unit32	KWH	Divide by 10 to get value
146	Solar Rectifier Total Power[2]	Unit32	KWH	Divide by 10 to get value
147	Solar Rectifier Total Power[3]	Unit32	KWH	Divide by 10 to get value
148	Solar Rectifier Total Power[3]	Unit32	KWH	Divide by 10 to get value
149	Solar Rectifier Total Power[4]	Unit32	KWH	Divide by 10 to get value
150	Solar Rectifier Total Power[4]	Unit32	KWH	Divide by 10 to get value
151	Solar Rectifier Total Power[5]	Unit32	KWH	Divide by 10 to get value
152	Solar Rectifier Total Power[5]	Unit32	KWH	Divide by 10 to get value
153	Solar Rectifier Total Power[6]	Unit32	KWH	Divide by 10 to get value
154	Solar Rectifier Total Power[6]	Unit32	KWH	Divide by 10 to get value
155	Solar Rectifier Total Power[7]	Unit32	KWH	Divide by 10 to get value
156	Solar Rectifier Total Power[7]	Unit32	KWH	Divide by 10 to get value
157	Solar Rectifier Total Power[8]	Unit32	KWH	Divide by 10 to get value
158	Solar Rectifier Total Power[8]	Unit32	KWH	Divide by 10 to get value
159	Solar Rectifier Total Power[9]	Unit32	KWH	Divide by 10 to get value
160	Solar Rectifier Total Power[9]	Unit32	KWH	Divide by 10 to get value
161	Solar Rectifier Total Power[10]	Unit32	KWH	Divide by 10 to get value
162	Solar Rectifier Total Power[10]	Unit32	KWH	Divide by 10 to get value
163	Solar Rectifier UP Time[1]	Unit32	Hour	Divide by 10 to get value

Reg. Address	Parameter	Data Type	Units	Remarks
164	Solar Rectifier UP Time[1]	Unit32	Hour	Divide by 10 to get value
165	Solar Rectifier UP Time[2]	Unit32	Hour	Divide by 10 to get value
166	Solar Rectifier UP Time[2]	Unit32	Hour	Divide by 10 to get value
167	Solar Rectifier UP Time[3]	Unit32	Hour	Divide by 10 to get value
168	Solar Rectifier UP Time[3]	Unit32	Hour	Divide by 10 to get value
169	Solar Rectifier UP Time[4]	Unit32	Hour	Divide by 10 to get value
170	Solar Rectifier UP Time[4]	Unit32	Hour	Divide by 10 to get value
171	Solar Rectifier UP Time[5]	Unit32	Hour	Divide by 10 to get value
172	Solar Rectifier UP Time[5]	Unit32	Hour	Divide by 10 to get value
173	Solar Rectifier UP Time[6]	Unit32	Hour	Divide by 10 to get value
174	Solar Rectifier UP Time[6]	Unit32	Hour	Divide by 10 to get value
175	Solar Rectifier UP Time[7]	Unit32	Hour	Divide by 10 to get value
176	Solar Rectifier UP Time[7]	Unit32	Hour	Divide by 10 to get value
177	Solar Rectifier UP Time[8]	Unit32	Hour	Divide by 10 to get value
178	Solar Rectifier UP Time[8]	Unit32	Hour	Divide by 10 to get value
179	Solar Rectifier UP Time[9]	Unit32	Hour	Divide by 10 to get value
180	Solar Rectifier UP Time[9]	Unit32	Hour	Divide by 10 to get value
181	Solar Rectifier UP Time[10]	Unit32	Hour	Divide by 10 to get value
182	Solar Rectifier UP Time[10]	Unit32	Hour	Divide by 10 to get value
183	Output Voltage Solar Charger [1]	Unit16	Volt	Divide by 10 to get value
184	Output Voltage Solar Charger [2]	Unit16	Volt	Divide by 10 to get value
185	Output Voltage Solar Charger [3]	Unit16	Volt	Divide by 10 to get value
186	Output Voltage Solar Charger [4]	Unit16	Volt	Divide by 10 to get value
187	Output Voltage Solar Charger [5]	Unit16	Volt	Divide by 10 to get value
188	Output Voltage Solar Charger [6]	Unit16	Volt	Divide by 10 to get value
189	Output Voltage Solar Charger [7]	Unit16	Volt	Divide by 10 to get value
190	Output Voltage Solar Charger [8]	Unit16	Volt	Divide by 10 to get value
191	Output Voltage Solar Charger [9]	Unit16	Volt	Divide by 10 to get value
192	Output Voltage Solar Charger [10]	Unit16	Volt	Divide by 10 to get value
193	Output Current Solar Charger [1]	Unit16	Amp	Divide by 10 to get value
194	Output Current Solar Charger [2]	Unit16	Amp	Divide by 10 to get value
195	Output Current Solar Charger [3]	Unit16	Amp	Divide by 10 to get value
196	Output Current Solar Charger [4]	Unit16	Amp	Divide by 10 to get value
197	Output Current Solar Charger [5]	Unit16	Amp	Divide by 10 to get value
198	Output Current Solar Charger [6]	Unit16	Amp	Divide by 10 to get value
199	Output Current Solar Charger [7]	Unit16	Amp	Divide by 10 to get value
200	Output Current Solar Charger [8]	Unit16	Amp	Divide by 10 to get value
201	Output Current Solar Charger [9]	Unit16	Amp	Divide by 10 to get value
202	Output Current Solar Charger [10]	Unit16	Amp	Divide by 10 to get value
203	Total Solar Charger Current	Unit16	Amp	Divide by 10 to get value
204	Reserved	UINT16		
205	Reserved	UINT16		
206	Reserved	UINT16		
207	Reserved	UINT16		

Reg. Address	Parameter	Data Type	Units	Remarks
208	Reserved	UINT16		
209	Reserved	UINT16		
210	Reserved	UINT16		
211	Reserved	UINT16		
212	Reserved	UINT16		
213	Reserved	UINT16		
214	Reserved	UINT16		
215	Reserved	UINT16		
216	Reserved	UINT16		
217	Reserved	UINT16		
218	Reserved	UINT16		
219	Reserved	UINT16		
220	Solar Charger Version Revision [1]	UINT32		
221	Solar Charger Version Revision [1]	UINT32		
222	Solar Charger Version Revision [2]	UINT32		
223	Solar Charger Version Revision [2]	UINT32		
224	Solar Charger Version Revision [3]	UINT32		
225	Solar Charger Version Revision [3]	UINT32		
226	Solar Charger Version Revision [4]	UINT32		
227	Solar Charger Version Revision [4]	UINT32		
228	Solar Charger Version Revision [5]	UINT32		
229	Solar Charger Version Revision [5]	UINT32		
230	Solar Charger Version Revision [6]	UINT32		
231	Solar Charger Version Revision [6]	UINT32		
232	Solar Charger Version Revision [7]	UINT32		
233	Solar Charger Version Revision [7]	UINT32		
234	Solar Charger Version Revision [8]	UINT32		
235	Solar Charger Version Revision [8]	UINT32		
236	Solar Charger Version Revision [9]	UINT32		
237	Solar Charger Version Revision [9]	UINT32		
238	Solar Charger Version Revision [10]	UINT32		
239	Solar Charger Version Revision [10]	UINT32		
240	Rectifier Version Revision [1]	UINT32		
241	Rectifier Version Revision [1]	UINT32		
242	Rectifier Version Revision [2]	UINT32		
243	Rectifier Version Revision [2]	UINT32		
244	Rectifier Version Revision [3]	UINT32		
245	Rectifier Version Revision [3]	UINT32		
246	Rectifier Version Revision [4]	UINT32		
247	Rectifier Version Revision [4]	UINT32		
248	Rectifier Version Revision [5]	UINT32		
249	Rectifier Version Revision [5]	UINT32		

Reg. Address	Parameter	Data Type	Units	Remarks
250	Rectifier Version Revision [6]	UINT32		
251	Rectifier Version Revision [6]	UINT32		
252	Rectifier Version Revision [7]	UINT32		
253	Rectifier Version Revision [7]	UINT32		
254	Rectifier Version Revision [8]	UINT32		
255	Rectifier Version Revision [8]	UINT32		
256	Rectifier Version Revision [9]	UINT32		
257	Rectifier Version Revision [9]	UINT32		
258	Rectifier Version Revision [10]	UINT32		
259	Rectifier Version Revision [10]	UINT32		
260	Rectifier Version Revision [11]	UINT32		
261	Rectifier Version Revision [11]	UINT32		
262	Rectifier Version Revision [12]	UINT32		
263	Rectifier Version Revision [12]	UINT32		
264	Rectifier Version Revision [13]	UINT32		
265	Rectifier Version Revision [13]	UINT32		
266	Rectifier Version Revision [14]	UINT32		
267	Rectifier Version Revision [14]	UINT32		
268	Rectifier Version Revision [15]	UINT32		
269	Rectifier Version Revision [15]	UINT32		
270	Reserved	UINT16		
271	Reserved	UINT16		
272	User Alarms [1 to 16]	UINT16		
273	User Alarms [17 to 32]	UINT16		
274	User Alarms [33 to 48]	UINT16		
275	User Alarms [49 to 64]	UINT16		
276	Group Alarms [1 to 16]	UINT16		
277	Group Alarms [17 to 32]	UINT16		
278	Group Alarms [33 to 48]	UINT16		
279	Group Alarms [49 to 64]	UINT16		
280	Analog Alarms [1 to 16]	UINT16		
281	Reserved	UINT16		
282	Reserved	UINT16		
283	Reserved	UINT16		
284	Reserved	UINT16		
285	Reserved	UINT16		
286	Reserved	UINT16		
287	Reserved	UINT16		

7.2 Alarm Register Description

Alarm Registers in above table give system status. A “1” in a bit location indicates presence of that alarm and “0” indicates a normal state. Following table gives details of all the alarm flags. Bit 0 is LSB and bit 15 is MSB.

Table 11- Alarm Descriptions

Sr. No.	Input Register	Bit Field	Description
1	Alarm Register 1	0	Reserved
2	Alarm Register 1	1	Any Phase Fail
3	Alarm Register 1	2	R Phase Fail
4	Alarm Register 1	3	Y Phase Fail
5	Alarm Register 1	4	B Phase Fail
6	Alarm Register 1	5	Mains Voltage High
7	Alarm Register 1	6	Mains Voltage Low
8	Alarm Register 1	7	Mains ON Battery Discharge
9	Alarm Register 1	8	Low Voltage Battery Disconnected
10	Alarm Register 1	9	DC Over Voltage
11	Alarm Register 1	10	DC Under Voltage
12	Alarm Register 1	11	System Over Load
13	Alarm Register 1	12	Temp Compensation Fail
14	Alarm Register 1	13	SPU Fail
15	Alarm Register 1	14	Battery Over Charge
16	Alarm Register 1	15	Battery 1 Isolated
17	Alarm Register 2	0	Battery 2 Isolated
18	Alarm Register 2	1	Battery 1 Reverse
19	Alarm Register 2	2	Battery 2 Reverse
20	Alarm Register 2	3	Battery 1 Fuse Fail
21	Alarm Register 2	4	Battery 2 Fuse Fail
22	Alarm Register 2	5	Rectifier Fail
23	Alarm Register 2	6	All Rectifier Fail
24	Alarm Register 2	7	All Solar HVLV Comm. Fail
25	Alarm Register 2	8	Solar Charger Fail
26	Alarm Register 2	9	All Solar Charger Fail
27	Alarm Register 2	10	Solar System Fail
28	Alarm Register 2	11	Battery Bank Discharge
29	Alarm Register 2	12	Battery 1 Discharge

Sr. No.	Input Register	Bit Field	Description
30	Alarm Register 2	13	Battery 2 Discharge
31	Alarm Register 2	14	Battery 1 First Float
32	Alarm Register 2	15	Battery 2 First Float
33	Alarm Register 3	0	AC Available
34	Alarm Register 3	1	Latched Alarm
35	Alarm Register 3	2	Charge Mode
36	Alarm Register 3	3	Boost Mode
37	Alarm Register 3	4	Boost Mode Flag1
38	Alarm Register 3	5	Boost Mode Flag2
39	Alarm Register 3	6	Manual Charge Mode
40	Alarm Register 3	7	BCL
41	Alarm Register 3	8	Power ON Start
42	Alarm Register 3	9	Power ON Start
43	Alarm Register 3	10	Battery 1 Test Status
44	Alarm Register 3	11	Battery 2 Test Status
45	Alarm Register 3	12	Battery1 Test Status Result
46	Alarm Register 3	13	Battery2 Test Status Result
47	Alarm Register 3	14	Battery Test Mode
48	Alarm Register 3	15	Conductance Test
49	Alarm Register 4	0	Battery 1 Test Fail
50	Alarm Register 4	1	Battery 2 Test Fail
51	Alarm Register 4	2	Reserved, Don't care
52	Alarm Register 4	3	Reserved, Don't care
53	Alarm Register 4	4	Reserved, Don't care
54	Alarm Register 4	5	Reserved, Don't care
55	Alarm Register 4	6	Reserved, Don't care
56	Alarm Register 4	7	Reserved, Don't care
57	Alarm Register 4	8	Reserved, Don't care
58	Alarm Register 4	9	Reserved, Don't care
59	Alarm Register 4	10	Reserved, Don't care

Sr. No.	Input Register	Bit Field	Description
60	Alarm Register 4	11	Reserved, Don't care
61	Alarm Register 4	12	Reserved, Don't care
62	Alarm Register 4	13	Reserved, Don't care
63	Alarm Register 4	14	Reserved, Don't care
64	Alarm Register 4	15	Reserved, Don't care
65	Rectifier Alarm Status	0	Rectifier 1 Fail
66	Rectifier Alarm Status	1	Rectifier 2 Fail
67	Rectifier Alarm Status	2	Rectifier 3 Fail
68	Rectifier Alarm Status	3	Rectifier 4 Fail
69	Rectifier Alarm Status	4	Rectifier 5 Fail
70	Rectifier Alarm Status	5	Rectifier 6 Fail
71	Rectifier Alarm Status	6	Rectifier 7 Fail
72	Rectifier Alarm Status	7	Rectifier 8 Fail
73	Rectifier Alarm Status	8	Rectifier 9 Fail
74	Rectifier Alarm Status	9	Rectifier 10 Fail
75	Rectifier Alarm Status	10	Rectifier 11 Fail
76	Rectifier Alarm Status	11	Rectifier 12 Fail
77	Rectifier Alarm Status	12	Rectifier 13 Fail
78	Rectifier Alarm Status	13	Rectifier 14 Fail
79	Rectifier Alarm Status	14	Rectifier 15 Fail
80	Rectifier Alarm Status	15	Rectifier 16 Fail
81	Rectifier Communication Status	0	Rectifier 1 Communication Error
82	Rectifier Communication Status	1	Rectifier 2 Communication Error
83	Rectifier Communication Status	2	Rectifier 3 Communication Error
84	Rectifier Communication Status	3	Rectifier 4 Communication Error
85	Rectifier Communication Status	4	Rectifier 5 Communication Error
86	Rectifier Communication Status	5	Rectifier 6 Communication Error
87	Rectifier Communication Status	6	Rectifier 7 Communication Error
88	Rectifier Communication Status	7	Rectifier 8 Communication Error
89	Rectifier Communication Status	8	Rectifier 9 Communication Error

Sr. No.	Input Register	Bit Field	Description
90	Rectifier Communication Status	9	Rectifier 10 Communication Error
91	Rectifier Communication Status	10	Rectifier 11 Communication Error
92	Rectifier Communication Status	11	Rectifier 12 Communication Error
93	Rectifier Communication Status	12	Rectifier 13 Communication Error
94	Rectifier Communication Status	13	Rectifier 14 Communication Error
95	Rectifier Communication Status	14	Rectifier 15 Communication Error
96	Rectifier Communication Status	15	Rectifier 16 Communication Error
97	Solar Charger Alarm Status	0	Solar Charger 1 Fail
98	Solar Alarm Status	1	Solar Charger 2 Fail
99	Solar Alarm Status	2	Solar Charger 3 Fail
100	Solar Alarm Status	3	Solar Charger 4 Fail
101	Solar Alarm Status	4	Solar Charger 5 Fail
102	Solar Alarm Status	5	Solar Charger 6 Fail
103	Solar Alarm Status	6	Solar Charger 7 Fail
104	Solar Alarm Status	7	Solar Charger 8 Fail
105	Solar Alarm Status	8	Solar Charger 9 Fail
106	Solar Alarm Status	9	Solar Charger 10 Fail
107	Solar Alarm Status	10	Solar Charger 11 Fail
108	Solar Alarm Status	11	Solar Charger 12 Fail
109	Solar Alarm Status	12	Solar Charger 13 Fail
110	Solar Alarm Status	13	Solar Charger 14 Fail
111	Solar Alarm Status	14	Solar Charger 15 Fail
112	Solar Alarm Status	15	Solar Charger 16 Fail
113	Solar Communication Status	0	Solar Charger 1 Communication Error
114	Solar Communication Status	1	Solar Charger 2 Communication Error
115	Solar Communication Status	2	Solar Charger 3 Communication Error
116	Solar Communication Status	3	Solar Charger 4 Communication Error
117	Solar Communication Status	4	Solar Charger 5 Communication Error
118	Solar Communication Status	5	Solar Charger 6 Communication Error
119	Solar Communication Status	6	Solar Charger 7 Communication Error

Sr. No.	Input Register	Bit Field	Description
120	Solar Communication Status	7	Solar Charger 8 Communication Error
121	Solar Communication Status	8	Solar Charger 9 Communication Error
122	Solar Communication Status	9	Solar Charger 10 Communication Error
123	Solar Communication Status	10	Solar Charger 11 Communication Error
124	Solar Communication Status	11	Solar Charger 12 Communication Error
125	Solar Communication Status	12	Solar Charger 13 Communication Error
126	Solar Communication Status	13	Solar Charger 14 Communication Error
127	Solar Communication Status	14	Solar Charger 15 Communication Error
128	Solar Communication Status	15	Solar Charger 16 Communication Error

7.3 User Alarm Register Description

The following register describes the bit location corresponding to its User Alarm status. There are total 64 User Alarms available in M2000 Controller which are divided into 4 registers of 16-bit each to show their status. Where “1” in a bit location indicates presence of that alarm and “0” indicates a normal state. Following table gives details of all the alarm flags. Bit 0 is LSB and bit 15 is MSB.

Register Number 272

B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
-----	-----	-----	-----	-----	-----	----	----	----	----	----	----	----	----	----	----

Bit	Description	Bit	Description
B0	User Alarm 1 Status	B8	User Alarm 9 Status
B1	User Alarm 2 Status	B9	User Alarm 10 Status
B2	User Alarm 3 Status	B10	User Alarm 11 Status
B3	User Alarm 4 Status	B11	User Alarm 12 Status
B4	User Alarm 5 Status	B12	User Alarm 13 Status
B5	User Alarm 6 Status	B13	User Alarm 14 Status
B6	User Alarm 7 Status	B14	User Alarm 15 Status
B7	User Alarm 8 Status	B15	User Alarm 16 Status

Register Number 273

B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
-----	-----	-----	-----	-----	-----	----	----	----	----	----	----	----	----	----	----

Bit	Description	Bit	Description
B0	User Alarm 16 Status	B8	User Alarm 24 Status
B1	User Alarm 17 Status	B9	User Alarm 25 Status
B2	User Alarm 18 Status	B10	User Alarm 26 Status
B3	User Alarm 19 Status	B11	User Alarm 27 Status
B4	User Alarm 20 Status	B12	User Alarm 28 Status
B5	User Alarm 21 Status	B13	User Alarm 29 Status
B6	User Alarm 22 Status	B14	User Alarm 30 Status
B7	User Alarm 23 Status	B15	User Alarm 31 Status

Register Number 274

B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
-----	-----	-----	-----	-----	-----	----	----	----	----	----	----	----	----	----	----

Bit	Description	Bit	Description
B0	User Alarm 33 Status	B8	User Alarm 41 Status
B1	User Alarm 34 Status	B9	User Alarm 42 Status
B2	User Alarm 35 Status	B10	User Alarm 43 Status
B3	User Alarm 36 Status	B11	User Alarm 44 Status
B4	User Alarm 37 Status	B12	User Alarm 45 Status
B5	User Alarm 38 Status	B13	User Alarm 46 Status
B6	User Alarm 39 Status	B14	User Alarm 47 Status
B7	User Alarm 40 Status	B15	User Alarm 48 Status

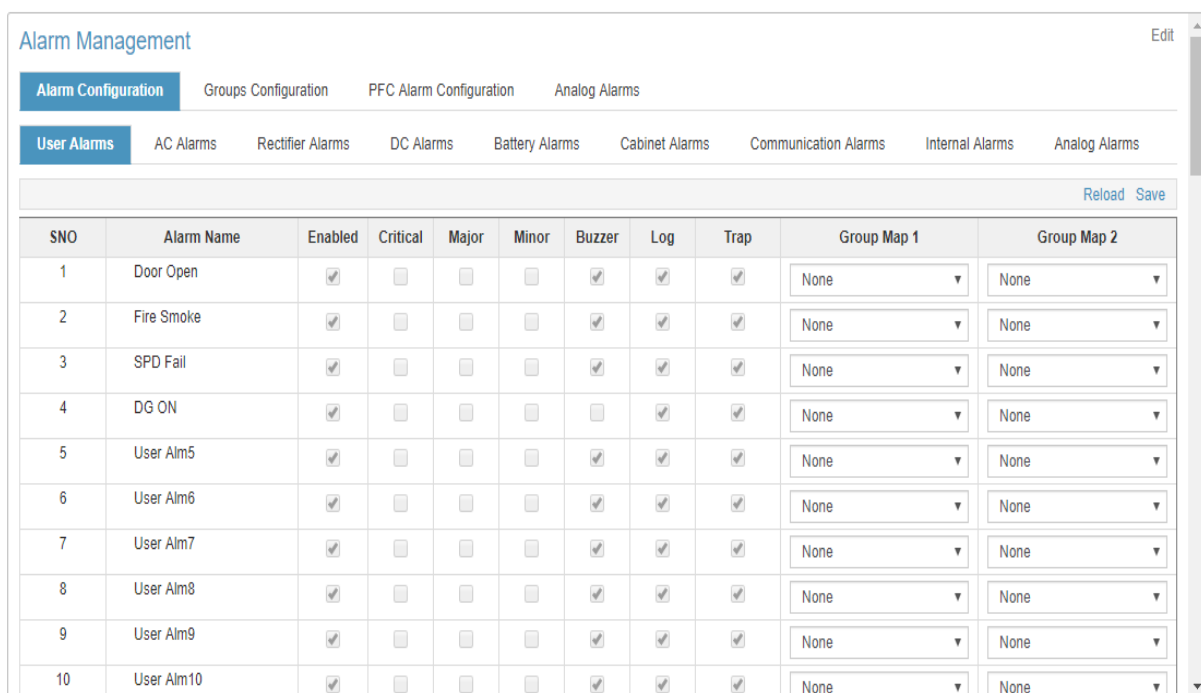
Register Number 275

B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
-----	-----	-----	-----	-----	-----	----	----	----	----	----	----	----	----	----	----

Bit	Description	Bit	Description
B0	User Alarm 49 Status	B8	User Alarm 57 Status
B1	User Alarm 50 Status	B9	User Alarm 58 Status
B2	User Alarm 51 Status	B10	User Alarm 59 Status

B3	User Alarm 52 Status	B11	User Alarm 60 Status
B4	User Alarm 53 Status	B12	User Alarm 61 Status
B5	User Alarm 54 Status	B13	User Alarm 62 Status
B6	User Alarm 55 Status	B14	User Alarm 63 Status
B7	User Alarm 56 Status	B15	User Alarm 64 Status

Please note that the User alarms might have been renamed as per system requirement in Configuration file. For correlating to configured user alarm name please refer web page under **Settings → Alarm Management → Alarm Configuration → User Alarms**. Following image relates the same.



SNO	Alarm Name	Enabled	Critical	Major	Minor	Buzzer	Log	Trap	Group Map 1	Group Map 2
1	Door Open	☒	☐	☐	☐	☒	☒	☒	None	None
2	Fire Smoke	☒	☐	☐	☐	☒	☒	☒	None	None
3	SPD Fail	☒	☐	☐	☐	☒	☒	☒	None	None
4	DG ON	☒	☐	☐	☐	☐	☒	☒	None	None
5	User Alm5	☒	☐	☐	☐	☒	☒	☒	None	None
6	User Alm6	☒	☐	☐	☐	☒	☒	☒	None	None
7	User Alm7	☒	☐	☐	☐	☒	☒	☒	None	None
8	User Alm8	☒	☐	☐	☐	☒	☒	☒	None	None
9	User Alm9	☒	☐	☐	☐	☒	☒	☒	None	None
10	User Alm10	☒	☐	☐	☐	☒	☒	☒	None	None

7.4 Group Alarm Register Description

The following register describes the bit location corresponding to its Group Alarm status. There are total 64 alarm groups available in M2000 Controller which are divided into 4 registers of 16-bit each to show their status. Where “1” in a bit location indicates presence of that alarm and “0” indicates a normal state. Following table gives details of all the alarm flags. Bit 0 is LSB and bit 15 is MSB.

Register Number 276

B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
-----	-----	-----	-----	-----	-----	----	----	----	----	----	----	----	----	----	----

Bit	Description	Bit	Description
B0	Group Alarm 1 Status	B8	Group Alarm 9 Status
B1	Group Alarm 2 Status	B9	Group Alarm 10 Status
B2	Group Alarm 3 Status	B10	Group Alarm 11 Status
B3	Group Alarm 4 Status	B11	Group Alarm 12 Status
B4	Group Alarm 5 Status	B12	Group Alarm 13 Status
B5	Group Alarm 6 Status	B13	Group Alarm 14 Status
B6	Group Alarm 7 Status	B14	Group Alarm 15 Status
B7	Group Alarm 8 Status	B15	Group Alarm 16 Status

Register Number 277

B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
-----	-----	-----	-----	-----	-----	----	----	----	----	----	----	----	----	----	----

Bit	Description	Bit	Description
B0	Group Alarm 16 Status	B8	Group Alarm 24 Status
B1	Group Alarm 17 Status	B9	Group Alarm 25 Status
B2	Group Alarm 18 Status	B10	Group Alarm 26 Status
B3	Group Alarm 19 Status	B11	Group Alarm 27 Status
B4	Group Alarm 20 Status	B12	Group Alarm 28 Status
B5	Group Alarm 21 Status	B13	Group Alarm 29 Status
B6	Group Alarm 22 Status	B14	Group Alarm 30 Status
B7	Group Alarm 23 Status	B15	Group Alarm 31 Status

Register Number 278

B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
-----	-----	-----	-----	-----	-----	----	----	----	----	----	----	----	----	----	----

Bit	Description	Bit	Description
B0	Group Alarm 33 Status	B8	Group Alarm 41 Status
B1	Group Alarm 34 Status	B9	Group Alarm 42 Status
B2	Group Alarm 35 Status	B10	Group Alarm 43 Status
B3	Group Alarm 36 Status	B11	Group Alarm 44 Status

Bit	Description	Bit	Description
B4	Group Alarm 37 Status	B12	Group Alarm 45 Status
B5	Group Alarm 38 Status	B13	Group Alarm 46 Status
B6	Group Alarm 39 Status	B14	Group Alarm 47 Status
B7	Group Alarm 40 Status	B15	Group Alarm 48 Status

Register Number 279

B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
-----	-----	-----	-----	-----	-----	----	----	----	----	----	----	----	----	----	----

Bit	Description	Bit	Description
B0	Group Alarm 49 Status	B8	Group Alarm 57 Status
B1	Group Alarm 50 Status	B9	Group Alarm 58 Status
B2	Group Alarm 51 Status	B10	Group Alarm 59 Status
B3	Group Alarm 52 Status	B11	Group Alarm 60 Status
B4	Group Alarm 53 Status	B12	Group Alarm 61 Status
B5	Group Alarm 54 Status	B13	Group Alarm 62 Status
B6	Group Alarm 55 Status	B14	Group Alarm 63 Status
B7	Group Alarm 56 Status	B15	Group Alarm 64 Status

Please note that the Group names might have been renamed as per system requirement in Configuration file. For correlating to configured group names please refer web page under All Group Alarms configuration can be seen or modified on web page under **Settings → Alarm Management → Group Configuration..**

Alarm Management
Edit

Alarm Configuration
Groups Configuration
PFC Alarm Configuration
Analog Alarms

No Of Groups 64

Group Names

GROUP1	Group1	GROUP2	Group2
GROUP3	Group3	GROUP4	Group4
GROUP5	Group5	GROUP6	Group6
GROUP7	Group7	GROUP8	Group8
GROUP9	Group9	GROUP10	Group10
GROUP11	Group11	GROUP12	Group12
GROUP13	Group13	GROUP14	Group14
GROUP15	Group15	GROUP16	Group16

7.5 Analog Alarm Register Description

The following registers describes the bit location corresponding to its Analog Alarm status. There are total 16 Analog Alarms available in M2000 Controller and there status are shown by a 16-bit register. Where “1” in a bit location indicates presence of that alarm and “0” indicates a normal state. Following table gives details of all the alarm flags. Bit 0 is LSB and bit 15 is MSB.

Register Number 280

B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
-----	-----	-----	-----	-----	-----	----	----	----	----	----	----	----	----	----	----

Bit	Description	Bit	Description
B0	Analog Alarm 1 Status	B8	Analog Alarm 9 Status
B1	Analog Alarm 2 Status	B9	Analog Alarm 10 Status
B2	Analog Alarm 3 Status	B10	Analog Alarm 11 Status
B3	Analog Alarm 4 Status	B11	Analog Alarm 12 Status
B4	Analog Alarm 5 Status	B12	Analog Alarm 13 Status
B5	Analog Alarm 6 Status	B13	Analog Alarm 14 Status
B6	Analog Alarm 7 Status	B14	Analog Alarm 15 Status
B7	Analog Alarm 8 Status	B15	Analog Alarm 16 Status

All Analog Alarms configuration can be seen or modified on web page under **Settings** → **Alarm Management** → **Analog Alarms** and the following image relates the same.

Alarm Management

Edit

Alarm Configuration

Groups Configuration

PFC Alarm Configuration

Analog Alarms

Alarm Names

Analog Alarm-1	Analog Alarm1	Analog Alarm-2	Analog Alarm2
Analog Alarm-3	Analog Alarm3	Analog Alarm-4	Analog Alarm4
Analog Alarm-5	Analog Alarm5	Analog Alarm-6	Analog Alarm6
Analog Alarm-7	Analog Alarm7	Analog Alarm-8	Analog Alarm8
Analog Alarm-9	Analog Alarm9	Analog Alarm-10	Analog Alarm10
Analog Alarm-11	Analog Alarm11	Analog Alarm-12	Analog Alarm12
Analog Alarm-13	Analog Alarm13	Analog Alarm-14	Analog Alarm14
Analog Alarm-15	Analog Alarm15	Analog Alarm-16	Analog Alarm16

----- End of Document -----